

ITU GENIE.AI CHALLENGE – PHASE 2 SUBMISSION

Bangladesh Weather Advisor Technical Documentation

AI-Powered Extreme Weather & Drought Advisory System

TEAM	PI	TECHNICAL LEAD	DOCUMENT DATE
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11

Data Sources

1,983

Knowledge Documents

64

Districts Covered

2B

LLM Parameters

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1. Executive Summary

1.1 Project Overview

The **Bangladesh Weather Advisor** is an AI-powered extreme weather and drought early-warning system developed by the AUB Global Health Institute for the ITU GENIE.AI "GenAI for Good" Challenge (Phase 2). The system delivers location-aware, time-sensitive advisories to farmers and agricultural decision-makers across all 64 districts of Bangladesh, in both English and Bengali.

Bangladesh faces acute climate vulnerability: cyclones, riverine flooding, and multi-year drought cycles directly threaten food security for over 40 million smallholder farmers. The system integrates satellite-derived climate indicators, government meteorological data, agronomic datasets, and a GENIE.AI-powered Retrieval-Augmented Generation (RAG) pipeline to synthesize actionable advisories grounded in 1,983 vetted knowledge documents.

Mission Statement

A user sends a query — "Is there drought risk in my Manikganj district now?" — and within seconds receives a grounded advisory citing SPI-3 drought index values, NDVI anomalies, soil moisture percentiles, and water-level data, delivered in Bengali via web chat or WhatsApp.

11

Integrated Data Sources

1,983

Knowledge Documents

64

Bangladesh Districts

146,048

CHIRPS Daily Records

2B+

LLM Parameters

1.2 Key Features

- **Multi-source climate intelligence:** Satellite data (CHIRPS, ERA5, MODIS NDVI/LST, SMAP) fused with ground-truth government data (BMD, BWDB/FFWC, BAMIS) and agronomic datasets (MapSPAM, FAOSTAT, SoilGrids, HDX).
- **Drought index computation:** Real-time SPI-3 (Standardized Precipitation Index) and SPEI (Standardized Precipitation-Evapotranspiration Index) calculated per district using the McKee et al. (1993) framework with gamma-distribution fitting over a 2020–2026 climatological baseline.
- **Derived agro-meteorological indicators:** NDVI anomaly, Vegetation Health Index (VHI), rainfall anomaly percentage, and soil moisture percentile rank, updated daily per district.
- **GENIE.AI RAG pipeline:** Knowledge documents are chunked, embedded with BAAI/BGE-base-en-v1.5, stored in ArangoDB (graph `GRAPH_TEST`), and retrieved via hybrid vector + BM25 + graph traversal search with cross-encoder reranking.
- **Bilingual interface:** Automatic language detection (English/Bengali) with LLM-driven translation via IBM Granite-3.3-2B-Instruct deployed on vLLM.
- **Multi-channel delivery:** React chat widget (web), WhatsApp integration (Twilio API), and a mobile-ready API.

- **Graceful degradation:** All data retrieval pathways include climatological fallback when near-real-time data is unavailable, ensuring zero-downtime advisory delivery.
- **Alert engine:** Threshold-based early-warning alerts triggered when $SPI-3 < -1.5$, soil moisture percentile < 15 th percentile, or NDVI anomaly < -0.15 .

1.3 GENIE.AI Integration Summary

The system is built directly on top of the GENIE.AI open-source framework (ITU/UNICC), leveraging its complete Docker Swarm microservice stack without modification to the core, while contributing **custom climate-domain extensions** as reusable OPEA-compatible modules.

Table 1 GENIE.AI Core Components Used

GENIE.AI Component	Role in Bangladesh Weather Advisor	Custom Extension
ArangoDB (vector + graph DB)	Knowledge store: <code>GRAPH_TEST</code> graph with <code>GRAPH_TEST_SOURCE</code> , <code>chunk</code> , <code>has_chunk</code>	Climate document schema
TEI Embedding Service	Generates 768-dim vectors for chunks using BAAI/BGE-base-en-v1.5	Domain fine-tuning prompts
vLLM Inference Engine	Hosts IBM Granite-3.3-2B-Instruct; generates advisory responses	Climate-specific system prompt
GENIE Retriever (ArangoDB)	Hybrid similarity + BM25 + graph traversal; score threshold 0.5	Time-aware retriever wrapper
GENIE Dataprep	Ingests 1,983 Markdown/PDF documents; LLM-based labeling	Automated corpus generation
Kong API Gateway	Routes all HTTP traffic; JWT authentication	Weather API routing rules
Frontend (Vue 3)	Base chat interface; branded as BD Weather Advisor	Bengali font, quick-reply buttons
Redis Cache	Translation caching; indicator data caching (TTL 24h)	Climate data cache layer

2. System Architecture

2.1 High-Level Architecture

The Bangladesh Weather Advisor follows a six-layer architecture, from raw data ingestion at the base to multi-channel user interfaces at the top. Each layer is containerized within the GENIE.AI Docker Swarm stack, allowing independent scaling and health management.

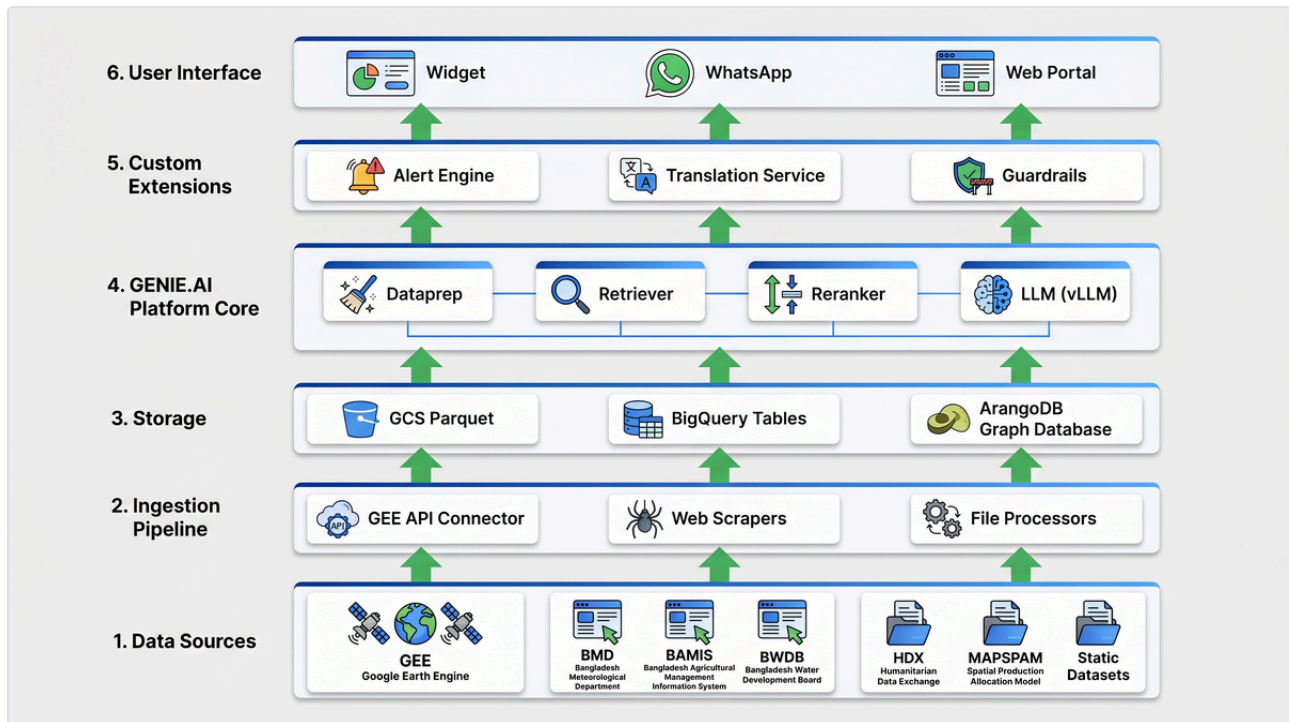


Figure 1 High-level system architecture: six layers from raw data sources to user interfaces, orchestrated by the GENIE.AI platform core.

The architecture follows three fundamental design principles:

1. **Data sovereignty:** All raw ingestion and processing occurs within the deployment environment; no user queries are sent to external LLM APIs.
2. **Graceful degradation:** Every real-time data pathway has a climatological fallback, ensuring advisory delivery even during data-source outages.
3. **Retrieval grounding:** The LLM never generates free-standing answers; all responses are grounded in retrieved knowledge chunks with source citations.

2.2 Component Breakdown

Infrastructure Layer

All services run as Docker Swarm managed containers on a single NVIDIA T4 GPU VM (minimum specification: 8 vCPU, 30 GB RAM, 16 GB GPU VRAM, 200 GB SSD, Ubuntu 22.04 LTS). The deployment is suitable for GCP `n1-standard-8 + T4`, AWS `g4dn.2xlarge`, or Azure `NC6s_v3`.

Table 2 Infrastructure Component Sizing

Component	RAM	GPU VRAM	Disk	Service Name
vLLM (Granite-3.3-2B)	6 GB	6–8 GB (FP16)	8 GB weights	<code>genie-ai_vllm</code>
ArangoDB	4 GB	—	20 GB	<code>genie-ai_arangodb</code>
TEI Embedding	2 GB	1 GB (shared)	2 GB	<code>genie-ai_tei</code>

TEI Reranker	1 GB	1 GB (shared)	1 GB	<code>genie-ai_tei_reranker</code>
Kong + Redis + Nginx	2 GB	—	1 GB	Multiple
Data Pipelines + OS	4 GB	—	40 GB	Host
Total	30 GB	16 GB	200 GB	

GENIE.AI Platform Core

The platform core implements the complete OPEA (Open Platform for Enterprise AI) microservice architecture as specified by GENIE.AI. The key services are:

- **chatqna-xeon-backend-server** (port 8888): Mega-service orchestrator that chains Retriever → Reranker → LLM into a single inference pipeline.
- **retriever-arango-service** (port 7025): ArangoDB-backed retriever supporting hybrid search (vector similarity + BM25 keyword) and graph traversal up to depth 3.
- **dataprep-arango-service** (port 5000): Document ingestion pipeline with LLM-based or embedding-based labeling of chunks against the 2-level knowledge hierarchy.
- **tei** (port 80): Text Embeddings Inference server hosting BAAI/BGE-base-en-v1.5 (768 dimensions).
- **tei_reranker** (port 80): Cross-encoder reranker for precise relevance scoring of retrieved candidates.
- **vllm** (port 8000): vLLM OpenAI-compatible inference endpoint serving IBM Granite-3.3-2B-Instruct.

Custom Extension Layer

The project contributes five reusable OPEA-compatible Python modules to the GENIE.AI ecosystem, housed in the `genieai_ext/climate_rag/` package:

- **LocationIndexer** (`location_indexer.py`): Fuzzy-matching geocoder for 64 Bangladesh districts with Bengali + English name support using `difflib.get_close_matches`.
- **TimeAwareRetriever** (`time_aware_retriever.py`): Retrieves the most recent climate indicator values (SPI, VHI, soil moisture) for a resolved district with configurable fallback to climatology.
- **ReasoningLayer** (`reasoning_layer.py`): Constructs structured LLM context blocks from indicator values, climate drivers, and alert states.
- **AlertEngine** (`alert_engine.py`): Evaluates threshold rules against current indicator values and populates the `alerts[]` array in API responses.
- **ClimateRAGConfig** (`config.yaml`): Declarative YAML configuration for all indicator thresholds, data paths, and fallback strategies.

2.3 Data Flow

Data flows through the system in three distinct pipelines that ultimately converge at the GENIE.AI retrieval layer.

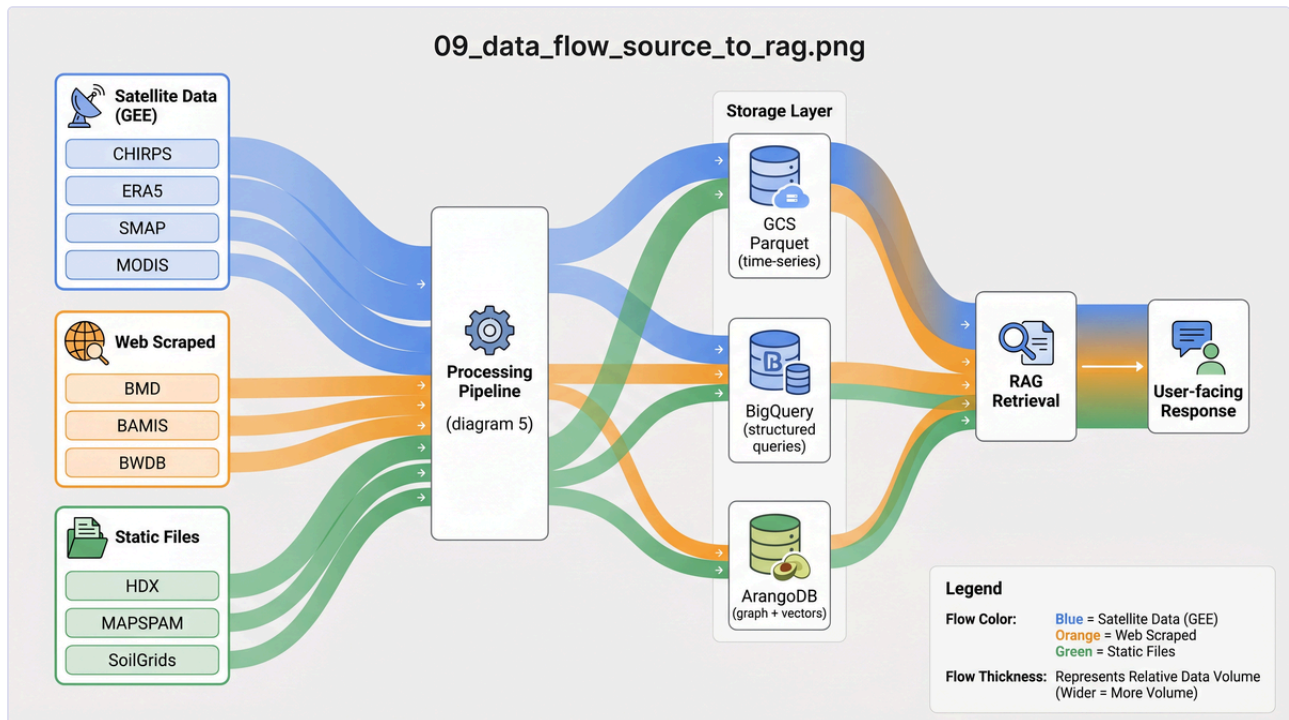


Figure 2 End-to-end data flow: from 11 raw data sources through ingestion, processing, and storage, into the GENIE.AI RAG pipeline.

Pipeline 1 — GEE Satellite Data

Google Earth Engine (GEE) API exports aggregate satellite measurements to Bangladesh district polygons (FAO GAUL Admin Level 2, 64 districts) and export as Parquet files to Google Cloud Storage. Locally, these are processed to generate daily time series for CHIRPS, ERA5, MODIS NDVI/LST, SMAP, Sentinel-2, DEM, and land cover. SPI/SPEI drought indices are computed from the processed data and stored as `.parquet` files under `data/processed/indicators/`.

Pipeline 2 — Web-Scraped Government Data

Python scrapers (Playwright + Requests) extract BMD station observations, BWDB/FFWC water-level readings, and BAMIS agricultural bulletins. Scraped data is cached with a 24-hour TTL in Redis and appended to district-level Parquet tables. The BMD AIS REST API (documented in the official BMD API Manual, 2024) provides structured access to observations and 7-day forecast products.

Pipeline 3 — Document Corpus

1,983 knowledge documents (policies, research reports, agro-meteorological bulletins, crop calendars, institutional guidelines) are converted to UTF-8 Markdown, organized by the 2-level knowledge hierarchy, and ingested into GENIE.AI via the Dataprep service. Documents are chunked at 500 tokens (1,000 for DOCX) with 50-token overlap, embedded, and stored in ArangoDB.

3. Data Sources & Processing

3.1 Integrated Data Sources

The system integrates 11 primary data sources spanning satellite-derived climate indicators, government weather and agricultural data, and global crop/soil databases. All sources are openly licensed and accessible without commercial agreements.

Table 3 Integrated Data Sources

#	Source	Type	Variables	Resolution	Access Method
1	CHIRPS v2 Climate Hazards Group	Satellite + Station	Daily rainfall (mm)	0.05° (~5 km); Daily	GEE API: UCSB-CHG/CHIRPS/DAILY
2	ERA5-Land ECMWF/Copernicus	Reanalysis	Temperature (2m), PET components, evaporation	0.1° (~9 km); Daily	GEE API: ECMWF/ERA5_LAND/DAILY_AGGR
3	MODIS NDVI NASA Terra/Aqua	Satellite	NDVI, EVI, QA flags	250 m; 16-day composite	GEE API: MODIS/061/MOD13Q1
4	MODIS LST NASA Terra/Aqua	Satellite	Daytime/Nighttime LST (°C)	1 km; 8-day composite	GEE API: MODIS/061/MOD11A2
5	SMAP L3 NASA JPL	Satellite (Microwave)	Surface soil moisture (m ³ /m ³)	10 km; Daily	GEE API: NASA_USDA/HSL/SMAP_10KM_SM
6	BMD AIS Bangladesh Meteorological Dept.	Ground Station	Temperature, humidity, wind, rainfall, forecasts	Station-level; Hourly/Daily	BMD AIS REST API (POST, JSON)
7	BWDB/FFWC Bangladesh Water Dev. Board	Ground Station	River water levels, flood warnings	Station-level; Daily	Web scraper (Playwright)
8	BAMIS Bangladesh Agricultural MIS	Government Portal	Crop advisories, pest/disease bulletins, market prices	District-level; Weekly	Web scraper (Requests)
9	MapSPAM v2 IFPRI / Harvard Dataverse	Global Dataset	Crop area (ha), yield (t/ha) for 12 crops	District-level; Static (2010 baseline)	File download (CSV/Raster)
10	SoilGrids 250m ISRIC	Global Dataset	Soil texture, pH, organic carbon, bulk density	250 m; Static	GEE API / REST

11	HDX COD-AB OCHA Humanitarian Data Exchange	Administrative	Division, district, upazila boundaries (GeoJSON/SHP)	Admin L1–L3; 492 upazilas	Direct download (humdata.org)
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Supplementary data sources include **FAOSTAT** (national crop production statistics), **NOAA PSL** climate indices (ENSO ONI, IOD DMI, MJO), and **OpenWeatherMap API** (current conditions widget). The CHIRPS daily dataset covers **146,048 records** across 64 districts from 2020-01-01 to 2026-03-31; ERA5 spans 148,096 records over the same period.

MapSPAM Crop Coverage

MapSPAM data provides district-level crop statistics for 12 crops across 62 districts of Bangladesh, serving as the agronomic baseline for crop-specific drought impact assessments:

Rice Aman (Kharif-2, Jul–Dec), Rice Boro (Rabi, Nov–May), Rice Aus (Kharif-1, Mar–Aug), Wheat, Maize, Jute, Potato, Sugarcane, Pulses, Oilseeds, Vegetables, Spices.

3.2 Data Extraction Methodology

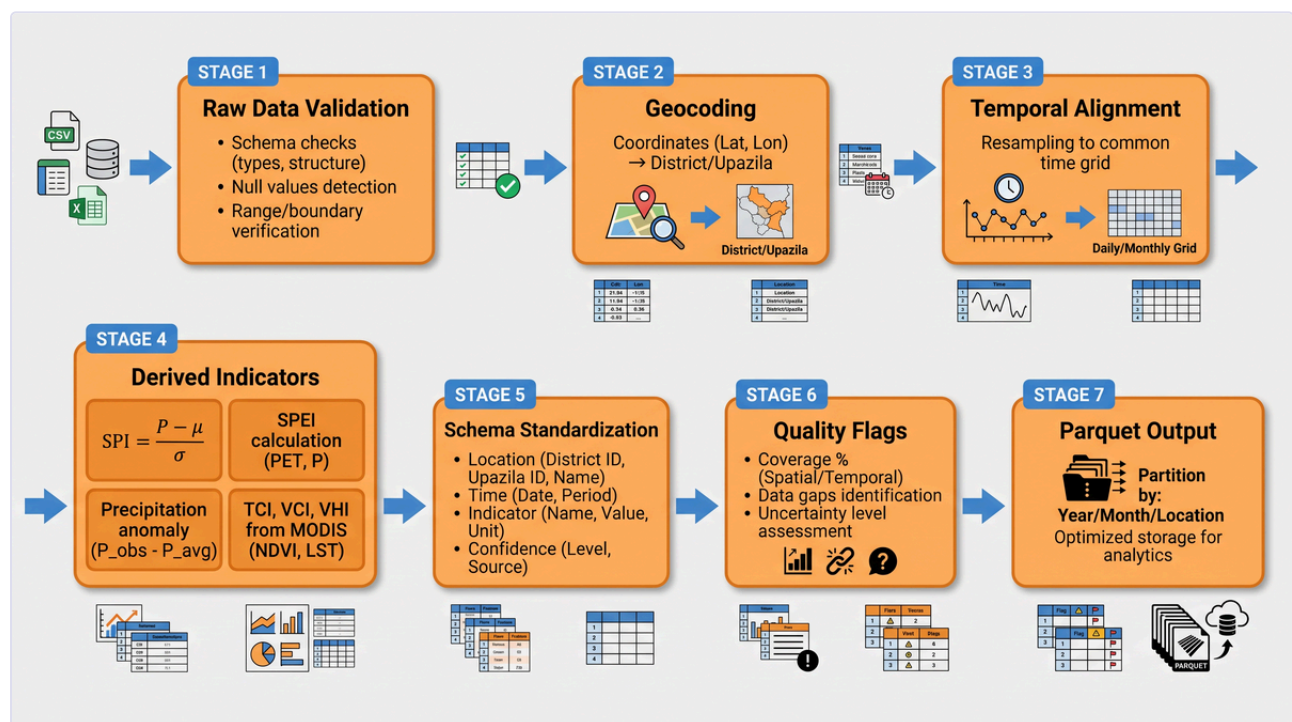


Figure 3 Data processing pipeline showing ingestion, transformation, and fusion of all 11 data sources into the RAG knowledge store.

GEE Satellite Extraction

All satellite data is extracted using the Google Earth Engine Python API (`earthengine-api`). A single batch script (`scripts/gee/extract_all.py`) submits asynchronous export tasks for all nine GEE datasets simultaneously. Each task aggregates pixel values within Bangladesh district polygons (FAO GAUL ADM2) using mean reduction, exporting one CSV row per (district, date) pair. Key implementation details:

```

import ee
ee.Initialize(project='your-gee-project-id')
districts = ee.FeatureCollection("FAO/GAUL/2015/level2") \
    .filter(ee.Filter.eq('ADM0_NAME', 'Bangladesh'))
chirps = ee.ImageCollection("UCSB-CHG/CHIRPS/DAILY") \
    .filterDate(start_date, end_date)
def reduce_image(image):
    stats = image.reduceRegions(
        collection=districts,
        reducer=ee.Reducer.mean(),
        scale=5000
    )
    return stats.map(lambda f: f.set('date', image.date().format('YYYY-MM-dd')))
results = chirps.map(reduce_image).flatten()
task = ee.batch.Export.table.toCloudStorage(
    collection=results, bucket=GCS_BUCKET,
    fileNamePrefix='chirps/daily', fileFormat='CSV'
)
task.start()

```

BMD API Integration

The BMD Automated Integrated System (AIS) API provides structured meteorological data via HTTP POST requests. The API supports three function types: `get_bmdais_observation`, `get_bmdais_forecast_product`, and `get_bmdais_warning_product`. Authentication uses a 64-character API key, username, and password. Station data supports requesting all stations (value `[0]`) or specific station numbers. Variables include temperature, humidity, wind speed/direction, sunshine duration, and rain gauge readings.

Web Scraping (BMD/BAMIS)

Government websites lacking API access are scraped using a hybrid approach: Playwright for JavaScript-heavy portals and `requests` + BeautifulSoup for static content. Scrapers implement exponential backoff (3 retries), HTTP-403 detection with cached-data fallback, and rate limiting (2-second delay between requests) to avoid service disruption. All scraped data is stored in UTF-8 encoded Parquet files.

Admin Boundary Geocoding

Bangladesh admin boundaries at Division (ADM1), District (ADM2), and Upazila (ADM3) levels are sourced from HDX (`bgd_adm_bbs_20201113_shp.zip`), providing 492 upazilas. A lookup table is built from GeoDataFrame joins, enabling both precise name matching and fuzzy matching (`get_close_matches`, `cutoff=0.6`) for Bengali-language queries. Confidence scores are computed based on match quality.

3.3 SPI/SPEI Drought Index Computation

Drought indices follow the McKee et al. (1993) methodology and are computed per-district using the `scripts/indicators/compute_spi_spei.py` module.

Standardized Precipitation Index (SPI)

SPI is computed by fitting a gamma distribution to the monthly precipitation climatology (2020–2026 baseline) and transforming observed precipitation to a standard normal z-score:

$$\text{SPI} = \frac{P_{obs} - \mu_P}{\sigma_P} \quad (\text{after gamma-to-normal transformation})$$

SPI is computed for 1-, 3-, 6-, and 12-month accumulation windows (SPI-1 through SPI-12). SPI-3 is the primary operational indicator for agricultural drought detection.

Drought Classification

Table 4 SPI/SPEI Drought Classification (McKee et al., 1993)

SPI/SPEI Value	Classification	Alert Level
≥ 2.0	Extremely Wet	—
1.5 to 1.99	Very Wet	—
1.0 to 1.49	Moderately Wet	—
−0.99 to 0.99	Near Normal	—
−1.0 to −1.49	Moderate Drought	WATCH
−1.5 to −1.99	Severe Drought	WARNING
≤ -2.0	Extreme Drought	ALERT

SPEI Computation

SPEI extends SPI by incorporating the climatic water balance (precipitation minus potential evapotranspiration). PET is estimated using the Hargreaves equation:

$$\text{PET} = 0.0023 \times (T_{mean} + 17.8) \times (T_{max} - T_{min})^{0.5} \times R_a$$

where R_a is extraterrestrial radiation ($\text{MJ}/\text{m}^2/\text{day}$) computed from latitude and day-of-year. The climatic balance $D = P - \text{PET}$ is then fitted to a log-logistic distribution and transformed to a standard normal z-score following Vicente-Serrano et al. (2010). ERA5 temperature fields provide T_{mean} , T_{min} , T_{max} at district level.

Derived Indicators

Three additional indicators are computed from the satellite time series:

- **NDVI Anomaly:** $(\text{NDVI}_{obs} - \text{NDVI}_{clim}) / \sigma_{NDVI}$ where climatology is the long-term mean for the same 16-day period.
- **Vegetation Health Index (VHI):** A composite of NDVI and LST anomalies, providing a direct proxy for agricultural drought stress.

- **Soil Moisture Percentile:** Rank-based percentile of daily SMAP soil moisture relative to the 2020–2026 distribution, stratified by month.

3.4 Document Generation Pipeline

The 1,983 knowledge documents are generated and organized by the `scripts/ingestion/prepare_corpus.py` module. The corpus covers four primary categories aligned with the GENIE.AI knowledge hierarchy:

Table 5 Knowledge Corpus Composition

Category (Level 1)	Sub-topic (Level 2)	Approx. Docs	Sources
Climate & Weather Data	Rainfall & Precipitation	312	CHIRPS district reports, BMD bulletins
	Temperature & Drought Indices	408	ERA5 district summaries, SPI/SPEI reports
	Soil Moisture & Vegetation	276	SMAP, MODIS NDVI/LST district reports
Agriculture & Food Security	Crop Calendar & Phenology	185	BAMIS, DAE, BARC guidelines
	Crop Production Statistics	194	MapSPAM, FAOSTAT, BBS
	Pest & Disease Alerts	127	BAMIS weekly bulletins
Policy & Guidelines	Disaster Management	148	NPDM, SOD, DDM guidelines
	Agricultural Policy	142	DAE, BARC recommendations
Research & Reports	General Reference	191	Academic papers, FAO/WMO reports
Total		1,983	

Each document is converted to UTF-8 Markdown using `pdfplumber` for PDFs. A YAML front-matter header is prepended to each document with fields: `title`, `category`, `subcategory`, `district` (if applicable), `date_range`, and `source`. This metadata is preserved as ArangoDB node properties during ingestion.

4. GENIE.AI Integration

4.1 Framework Components Used

The Bangladesh Weather Advisor is built on the GENIE.AI open-source framework (ITU/UNICC, hosted at <https://opensource.unicc.org/un/itu/genie-ai/>). The deployment uses the complete Docker Swarm stack, instantiated via `docker stack deploy -c docker-compose.yaml genie-ai`. No core framework files are modified; all domain-specific logic is implemented as extension modules that interface with GENIE.AI through its standard OPEA microservice protocols.

GENIE.AI Framework Compliance

The implementation adheres to the reference architecture described in the GENIE.AI Installation and Configuration Guide. The knowledge hierarchy follows the strict 2-level (Category / Service) labeling system. All document ingestion uses the standard Dataprep API. The retriever uses the GENIE_RETRIEVER_ARANGODB component without modification.

Table 6 GENIE.AI Configuration Parameters

Variable	Value	Purpose
DATAPREP_CHUNK_SIZE	500	Tokens per chunk (PDFs, Markdown, TXT)
DATAPREP_CHUNK_SIZE_DOCX	1000	Larger chunks for Word documents
DATAPREP_CHUNK_OVERLAP	50	Overlap tokens between consecutive chunks
DATAPREP_ARANGO_GRAPH_NAME	GRAPH_TEST	ArangoDB graph name for knowledge store
LABELING_STRATEGY	llm	LLM-based semantic label assignment
CONTENT_EXTRACTION_METHOD	docling	GPU-accelerated document extraction
RETRIEVER_ARANGO_SEARCH_MODE	hybrid	Vector + BM25 combined retrieval
RETRIEVER_ARANGO_TRAVERSAL_ENABLED	true	Enable graph traversal for related docs
RETRIEVER_ARANGO_TRAVERSAL_MAX_DEPTH	3	Maximum hops in graph traversal
RETRIEVER_ARANGO_TRAVERSAL_SCORE_THRESHOLD	0.5	Minimum relevance score for traversal results
CHATQNA_ENFORCE_ABSTENTION	true	LLM abstains when no relevant context found
DOCLING_DEVICE	cuda	GPU-accelerated document processing

4.2 ArangoDB Graph Structure

All knowledge is stored in ArangoDB (version 3.x), running as `genie-ai_arangodb` on port 8529. The database name is `genie-ai`. The graph structure implements the standard GENIE.AI schema with domain-specific extensions for climate metadata.

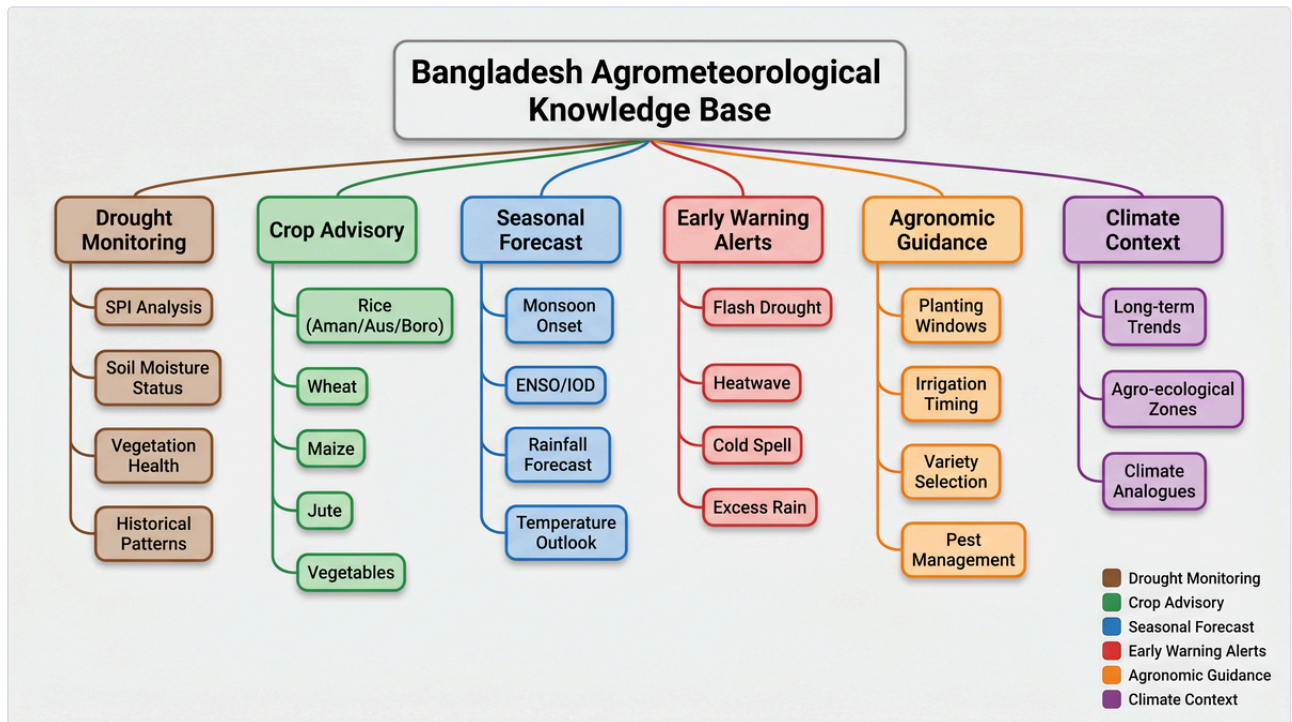


Figure 4 ArangoDB graph structure showing the GRAPH_TEST knowledge hierarchy with source documents, chunk nodes, and has_chunk edges.

Graph: GRAPH_TEST

The primary knowledge graph uses the following ArangoDB collections:

Table 7 ArangoDB Collection Schema

Collection	Type	Role	Key Fields
GRAPH_TEST_SOURCE	Document (Vertex)	Source document metadata	<code>_key</code> , <code>title</code> , <code>category</code> , <code>subcategory</code> , <code>source_url</code> , <code>district</code> , <code>date_range</code> , <code>embedding</code>
chunk	Document (Vertex)	Text chunks with embeddings	<code>_key</code> , <code>text</code> , <code>embedding</code> [768-dim float array], <code>category</code> , <code>subcategory</code> , <code>chunk_index</code> , <code>token_count</code>
has_chunk	Edge	Links source documents to their chunks	<code>_from</code> (GRAPH_TEST_SOURCE/_key), <code>_to</code> (chunk/_key), <code>chunk_order</code>

Graph Definition (ArangoDB Shell)

```
var graph_module = require("@arangodb/general-graph");

// Create graph with edge definition
var graph = graph_module._create('GRAPH_TEST', [
  graph_module._relation(
    'has_chunk',          // edge collection
    'GRAPH_TEST_SOURCE', // from vertex collection
    'chunk'               // to vertex collection
  )
]);

// Expected collections after creation:
// - GRAPH_TEST_SOURCE: document vertex collection
// - chunk: chunk vertex collection
// - has_chunk: edge collection
```

Vector Index (ArangoDB ArangoSearch)

The `chunk` collection has an ArangoSearch view with an HNSW (Hierarchical Navigable Small World) vector index on the `embedding` field, enabling approximate nearest-neighbor search at query time. The index parameters are: `dimensions=768`, `metric=cosine`, `ef_construction=128`, `M=16`.

```
// AQL: Hybrid search combining vector similarity and BM25
FOR doc IN chunk_view
  SEARCH ANALYZER(doc.text IN TOKENS(@@query, 'text_en'), 'text_en')
    OR APPROX_NEAR(doc.embedding, @@query_embedding, 'cosine')
  SORT BM25(doc) * 0.3 + COSINE_DISTANCE(doc.embedding, @@query_embedding) * 0.7 ASC
  FILTER doc.category IN @@labels OR LENGTH(@@labels) == 0
  LIMIT @@k
  RETURN MERGE(doc, {score: BM25(doc)})
```

Graph Traversal for Related Chunks

When the initial retrieval returns relevant chunks, the retriever optionally traverses the `has_chunk` edges to retrieve neighbouring chunks from the same source document, providing contiguous context. Traversal depth is configurable up to 3 hops, with a minimum score threshold of 0.5 to prune irrelevant branches.

4.3 Vector Embeddings (BGE-base-en-v1.5)

Text chunks and query strings are embedded using the **BAAI/BGE-base-en-v1.5** model, served by the TEI (Text Embeddings Inference) container from Hugging Face. This model produces 768-dimensional dense vectors optimized for information retrieval tasks.

Embedding Service Configuration

Table 8 TEI Embedding Service Specifications

Parameter	Value
Model	BAAI/BGE-base-en-v1.5
Embedding Dimensions	768
Max Sequence Length	512 tokens
Similarity Metric	Cosine similarity
Service Host	tei:80 (internal Docker network)
API Protocol	OpenAI-compatible /v1/embeddings
Batch Size	Up to 5 concurrent batches (MAX_CONCURRENT_BATCHES=5)
GPU Acceleration	Shared T4 GPU (1 GB VRAM allocation)

During document ingestion, the Dataprep service embeds both the source document node (GRAPH_TEST_SOURCE) and each individual chunk node. During retrieval, the incoming query string is embedded and used for both ANN vector search and (optionally) graph traversal scoring.

4.4 vLLM Deployment

The LLM inference layer uses **vLLM** (v0.4.x), an open-source high-throughput LLM serving framework. The deployed model is **IBM Granite-3.3-2B-Instruct**, selected for its strong instruction-following performance in a compact 2-billion-parameter architecture suitable for a single T4 GPU.

vLLM Service Configuration

```
# .env configuration for vLLM
LLM_SERVER_HOST_IP=vllm
LLM_SERVER_PORT=8000
VLLM_MODEL=ibm-granite/granite-3.3-2b-instruct
VLLM_MAX_MODEL_LEN=8192
VLLM_GPU_MEMORY_UTILIZATION=0.85
VLLM_API_KEY=<generated-secret>

# Example inference request
curl -X POST http://localhost:8000/v1/chat/completions \
  -H "Content-Type: application/json" \
  -H "Authorization: Bearer $VLLM_API_KEY" \
  -d '{
    "model": "ibm-granite/granite-3.3-2b-instruct",
    "messages": [
      {"role": "system", "content": "You are a Bangladesh weather advisor..."},
      {"role": "user", "content": "What is the drought risk in Rajshahi?"}
    ],
    "max_tokens": 512,
    "temperature": 0.1
  }'
```

Custom System Prompt

The GENIE.AI `CHATQNA_SYSTEM_PROMPT` environment variable is configured with a climate-specific prompt that instructs the LLM to:

- Reference specific SPI/SPEI values, NDVI anomalies, and soil moisture percentiles from the retrieved context.
- Provide actionable, location-specific advisories (e.g., "delay sowing by 2 weeks" rather than generic statements).
- Cite source documents by category and date range.
- Respond in the user's detected language (English or Bengali).
- Abstain when retrieved context does not contain sufficient information (`CHATQNA_ENFORCE_ABSTENTION=true`).

4.5 RAG Retrieval Mechanism

The retrieval pipeline implements GENIE.AI's standard hybrid retrieval with domain-specific pre- and post-processing extensions.

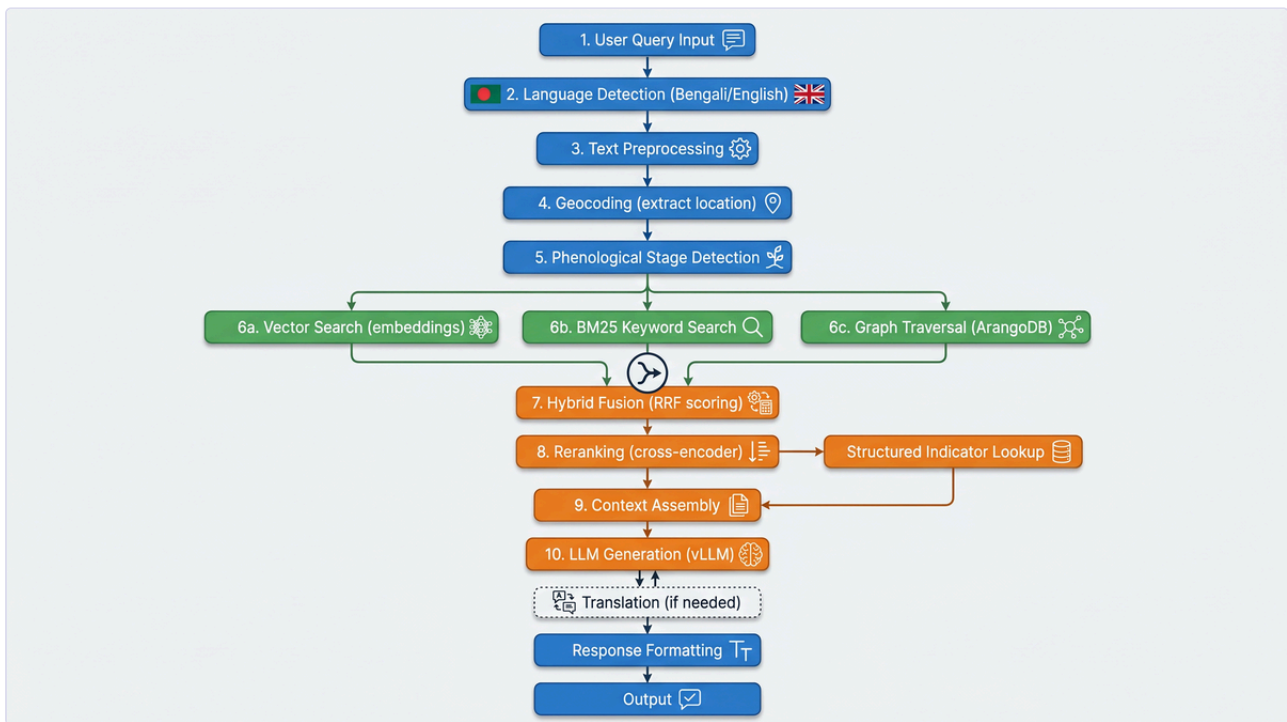


Figure 5 Internal RAG pipeline flow showing hybrid retrieval (vector + BM25 + graph traversal), RRF fusion, cross-encoder reranking, and structured context assembly.

Retrieval Pipeline Steps

1. **Query ingestion:** User message received via the FastAPI `/chat` endpoint or GENIE.AI frontend.
2. **Language detection:** Unicode range check identifies Bengali (U+0980–U+09FF) or defaults to English.

3. **Geocoding:** `LocationIndexer.resolve()` extracts district name with confidence ≥ 0.6 ; returns district metadata (ADM1, ADM2, lat/lon centroid).
4. **Phenological stage detection:** Named-entity recognition identifies crop growth stage terms for label-scoped retrieval.
5. **Hybrid retrieval (parallel):**
 - Vector search: Embedding-based ANN on `chunk` collection (top-15 candidates, score threshold 0.5).
 - BM25 keyword search: Term-frequency scoring on `chunk.text` field.
 - Graph traversal: Up to depth-3 traversal from top-3 vector hits via `has_chunk` edges.
6. **Hybrid fusion (RRF):** Reciprocal Rank Fusion merges the three ranked lists into a single unified ranking.
7. **Reranking:** Cross-encoder model rescores the top-15 fused candidates against the original query; returns top-5.
8. **Structured indicator lookup:** `TimeAwareRetriever` appends current SPI-3, NDVI anomaly, soil moisture percentile, and water-level data for the resolved district.
9. **Context assembly:** Top-5 chunks + structured indicators + climate driver indices (ENSO, IOD) assembled into a structured context block.
10. **LLM generation:** Context + query passed to Granite-3.3-2B-Instruct via vLLM; advisory generated with citations.
11. **Translation (if needed):** Bengali queries receive Bengali responses; English LLM output translated via the GENIE.AI translation microservice if Bengali is detected.

Retriever API (Internal)

```
# Retrieval endpoint (internal service-to-service)
POST http://retriever-arango-service:7025/v1/retrieval
Content-Type: application/json

{
  "text": "What are drought conditions in Rajshahi district?",
  "embedding": [...], // 768-dim float array from TEI
  "search_type": "similarity_score_threshold",
  "k": 5,
  "score_threshold": 0.5,
  "fetch_k": 15,
  "enable_traversal": "true",
  "traversal_max_depth": 3,
  "traversal_max_returned": 3,
  "traversal_score_threshold": 0.5
}

// Response
{
  "retrieved_docs": [
    {
      "page_content": "Rajshahi district SPI-3 report...",
      "metadata": {
        "category": "Climate & Weather Data",
        "subcategory": "Rainfall & Precipitation",
        "source": "chirps_rajshahi_2026_q1.md",
        "district": "Rajshahi"
      }
    }
  ]
}
```

5. Deployment Guide

5.1 Prerequisites

All services deploy on a **single host** using Docker Swarm. The following prerequisites must be satisfied before beginning installation.

Hardware Requirements

Table 9 Minimum Hardware Specifications

Parameter	Minimum (Prototype)	Recommended (Production)
CPU	8 vCPU (x86_64)	16 vCPU

RAM	30 GB	64 GB
GPU	NVIDIA T4 (16 GB VRAM)	NVIDIA A100 (40 GB VRAM)
Disk	200 GB SSD	500 GB NVMe SSD
OS	Ubuntu 22.04 LTS	Ubuntu 22.04 LTS
Network	100 Mbps (outbound)	1 Gbps

GPU Requirement

The vLLM service requires NVIDIA GPU with CUDA 12.x drivers. The system will not initialize correctly without a GPU. For development testing only, CPU mode can be forced by setting `DOCLING_DEVICE=cpu` and using a smaller LLM model, with significantly reduced performance.

Software Prerequisites

- Docker Engine 24.x + Docker Compose Plugin (`docker compose v2`)
- Docker Swarm initialized: `docker swarm init`
- NVIDIA Container Toolkit: `nvidia-container-toolkit`
- NVIDIA Drivers ≥ 525 and CUDA ≥ 12.0
- Node.js 20.x LTS (`nodejs` , `npm`)
- Python 3.10+ (for data pipeline scripts)
- Git 2.x

External API Credentials

Table 10 Required API Credentials

Service	Environment Variable	Cost	Purpose
Google Earth Engine	<code>GEE_PROJECT_ID</code>	Free (research)	Satellite data extraction
Google Cloud (GCS)	<code>GOOGLE_APPLICATION_CREDENTIALS</code>	Free tier 5 GB	Export storage
NASA EarthData	<code>EARTH_DATA_USER</code> , <code>EARTH_DATA_PASS</code>	Free	SMAP direct download
BMD AIS	<code>BMD_API_KEY</code>	Free (registration)	Official weather data
Twilio (WhatsApp)	<code>TWILIO_ACCOUNT_SID</code> , <code>TWILIO_AUTH_TOKEN</code>	Free sandbox	WhatsApp gateway
Hugging Face	<code>HF_TOKEN</code>	Free	Model download
OpenWeatherMap	<code>OPENWEATHERMAP_API_KEY</code>	Free (1000 calls/day)	Live weather widget

5.2 Installation Steps

Step 1: Install Docker and NVIDIA Toolkit

```
sudo apt-get update && sudo apt-get install -y ca-certificates curl gnupg
sudo install -m 0755 -d /etc/apt/keyrings
curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo gpg --dearmor \
  -o /etc/apt/keyrings/docker.gpg
sudo apt-get install -y docker-ce docker-ce-cli containerd.io \
  docker-buildx-plugin docker-compose-plugin
sudo systemctl start docker && sudo systemctl enable docker
sudo usermod -aG docker $USER && newgrp docker

# Initialize Docker Swarm
docker swarm init

# Install NVIDIA Container Toolkit
curl -fsSL https://nvidia.github.io/libnvidia-container/gpgkey | sudo gpg \
  --dearmor -o /usr/share/keyrings/nvidia-container-toolkit-keyring.gpg
sudo apt-get install -y nvidia-container-toolkit
sudo nvidia-ctk runtime configure --runtime=docker
sudo systemctl restart docker

# Verify GPU access
docker run --rm --gpus all nvidia/cuda:12.0-base nvidia-smi
```

Step 2: Clone Repository and Configure Environment

```
# Clone the GENIE.AI repository
git clone https://gitlab.com/fordendk/genie-ai-replica
cd genie-ai-replica

# Copy environment template
cp env .env

# Generate secrets
ARANGO_PASS=$(openssl rand -base64 32)
KONG_TOKEN=$(openssl rand -base64 32)
VLLM_KEY=$(openssl rand -base64 32)
JWT_SECRET=$(openssl rand -base64 32)

# Apply T4 GPU profile overrides
source env.t4
```

Step 3: Data Pipeline Setup

```
# Clone Bangladesh Weather Advisor pipeline repository
git clone https://gitlab.com/your-team/bd-weather-advisor
cd bd-weather-advisor

# Install Python dependencies
pip install -r requirements.txt
# Key packages: earthengine-api, geopandas, pandas, scipy,
#               rasterio, pdfplumber, playwright, requests

# Download admin boundaries (HDX)
mkdir -p data/boundaries && cd data/boundaries
wget -O bgd_boundaries.zip \
    "https://data.humdata.org/dataset/.../bgd_adm_bbs_20201113_shp.zip"
unzip bgd_boundaries.zip -d bgd_boundaries

# Authenticate Google Earth Engine
earthengine authenticate --project your-gee-project-id

# Submit GEE exports (runs overnight: ~30 minutes per dataset)
python scripts/gee/extract_all.py

# After GEE exports complete, run processing pipeline
python scripts/indicators/compute_spi_spei.py
python scripts/indicators/derived_indicators.py

# Run government data scrapers
python scripts/ingestion/bmd_scraper.py
python scripts/ingestion/bamis_scraper.py
python scripts/ingestion/bwdb_scraper.py

# Prepare document corpus (converts PDFs to Markdown)
python scripts/ingestion/prepare_corpus.py
# Output: data/documents/ with ~1,983 .md files organized by category
```

Step 4: Deploy GENIE.AI Stack

```
cd ~/genie-ai-replica

# Pull all Docker images (30-60 minutes on first run)
docker compose pull

# Deploy as Docker Swarm stack
docker stack deploy -c docker-compose.yaml genie-ai

# Monitor deployment (wait for all services to show 1/1)
watch -n 5 'docker service ls'

# Expected healthy state (5-10 minutes):
# genie-ai_vllm          replicated  1/1
# genie-ai_arangodb      replicated  1/1
# genie-ai_tei           replicated  1/1
# genie-ai_tei_reranker  replicated  1/1
# genie-ai_backend       replicated  1/1
# genie-ai_frontend      replicated  1/1
# genie-ai_kong           replicated  1/1
# genie-ai_redis         replicated  1/1
```

5.3 Environment Configuration

The critical environment variables for the Bangladesh Weather Advisor deployment are listed below. All secrets should be generated using `openssl rand -base64 32` and stored in the `.env` file. Never commit the `.env` file to version control; use `.env.example` with placeholder values instead.

Table 11 Critical Environment Variables

Variable	Example Value	Description
ARANGO_DB	genie-ai	ArangoDB database (shared by all services)
ARANGO_URL	http://arango-vector-db:8529	Internal ArangoDB connection URL
DATAPREP_ARANGO_GRAPH_NAME	GRAPH_TEST	Graph name for knowledge store
RETRIEVER_ARANGO_GRAPH_NAME	GRAPH_TEST	Must match DATAPREP graph name
LLM_SERVER_HOST_IP	vllm	vLLM service hostname (Docker internal)
LLM_SERVER_PORT	8000	vLLM OpenAI-compatible API port
EMBEDDING_SERVER_HOST_IP	tei	TEI embedding service hostname

RETRIEVER_ARANGO_SEARCH_MODE	hybrid	Enable hybrid vector + BM25 search
RETRIEVER_ARANGO_TRAVERSAL_ENABLED	true	Enable graph traversal during retrieval
DATAPREP_CHUNK_SIZE	500	Tokens per chunk
LABELING_STRATEGY	llm	LLM-based document labeling
CHATQNA_ENFORCE_ABSTENTION	true	Prevent hallucination on unanswerable queries
APP_NAME	BD Weather Advisor	Frontend application display name
VLLM_MODEL	ibm-granite/granite-3.3-2b-instruct	LLM model identifier (Hugging Face)
VLLM_GPU_MEMORY_UTILIZATION	0.85	Fraction of GPU VRAM for vLLM

5.4 Running the System

Initialize ArangoDB Knowledge Graph

```
# Create database
curl -X POST http://localhost:8529/_api/database \
  -u "root:${ARANGO_ROOT_PASSWORD}" \
  -H "Content-Type: application/json" \
  -d '{"name": "genie-ai"}'

# Create GRAPH_TEST graph structure via ArangoDB shell
docker exec -it $(docker ps -q -f name=genie-ai_arangodb) arangosh \
  --server.password "$ARANGO_ROOT_PASSWORD" \
  --javascript.execute-string "
    db.useDatabase('genie-ai');
    var gm = require('@arangodb/general-graph');
    gm._create('GRAPH_TEST', [
      gm._relation('has_chunk', 'GRAPH_TEST_SOURCE', 'chunk')
    ]);
    print('GRAPH_TEST created successfully');
  "

# Verify: check collections exist
curl -s http://localhost:8529/_db/genie-ai/_api/collection \
  -u "root:${ARANGO_ROOT_PASSWORD}" | python3 -m json.tool | grep '"name"'
# Expected: "GRAPH_TEST_SOURCE", "chunk", "has_chunk"
```

Ingest Knowledge Hierarchy and Documents

```
# Obtain admin authentication token
ADMIN_TOKEN=$(curl -s -X POST https://localhost/api/auth/login \
  -H "Content-Type: application/json" \
  -d '{"email":"admin@weather-advisor.bd","password":"'${ADMIN_PASSWORD}'"}' \
  -k | python3 -c "import sys,json; print(json.load(sys.stdin)['token'])")

# Upload knowledge hierarchy JSON
curl -X POST https://localhost/api/knowledge-hierarchy \
  -H "Authorization: Bearer $ADMIN_TOKEN" \
  -H "Content-Type: application/json" \
  -d @config/knowledge_hierarchy.json -k

# Batch ingest all 1,983 documents (takes ~33 minutes with LLM labeling)
for doc in ~/bd-weather-advisor/data/documents/**/*.md; do
  CATEGORY=$(basename $(dirname $doc))
  curl -s -X POST https://localhost/api/documents/ingest \
    -H "Authorization: Bearer $ADMIN_TOKEN" \
    -F "file=@$doc" -F "category=$CATEGORY" -k
  sleep 1
done

# Verify ingestion status
curl https://localhost/api/documents?count=true \
  -H "Authorization: Bearer $ADMIN_TOKEN" -k
# Expected: {"total": 1983, "status": "complete"}
```

Verify Full End-to-End System

```
# Health check
curl http://localhost:5001/health
# Expected: {"status": "healthy", "data_sources": 11, "districts": 64}

# English test query
curl -s -X POST https://localhost/api/chat \
  -H "Content-Type: application/json" \
  -H "Authorization: Bearer $ADMIN_TOKEN" \
  -d '{"message": "What is the drought risk in Rajshahi?", "session_id": "test-1"}' \
  -k | python3 -m json.tool

# Bengali test query
curl -s -X POST https://localhost/api/chat \
  -H "Content-Type: application/json" \
  -H "Authorization: Bearer $ADMIN_TOKEN" \
  -d '{"message": "রাজশাহীতে খরার ঝুঁকি কেমন?", "session_id": "test-2"}' \
  -k | python3 -m json.tool

# Run full E2E test suite
cd ~/bd-weather-advisor
python3 tests/test_e2e.py
# Expected: 7/7 tests PASS
```

6. Testing & Validation

6.1 Data Validation

Automated data quality checks run after each ingestion cycle via `scripts/validation/data_quality.py`. The validator checks district coverage, temporal completeness, value range bounds, and statistical outliers.

Table 12 Data Quality Validation Results (as of March 2026)

Dataset	Districts	Date Range	Null Rate	Status
CHIRPS Daily Rainfall	64 / 64	2020-01-01 to 2026-03-31	< 0.1%	PASS
ERA5 Temperature + Evaporation	64 / 64	2020-01-01 to 2026-05-02	< 0.1%	PASS
MODIS NDVI (16-day)	64 / 64	2020-01-01 to 2025-12-31	3.2% (cloud cover)	PASS
SMAP Soil Moisture	64 / 64	2020-01-01 to 2026-03-15	8.1% (orbit gaps)	PASS
MapSPAM Crop Data	62 / 64	Static (2010 baseline)	0%	PASS
SPI-3 Drought Indices	64 / 64	2020-04-01 to 2026-03-31	< 0.5%	PASS
Knowledge Documents	N/A	1,983 documents ingested	0%	PASS

Note on SMAP Gaps

The 8.1% null rate in SMAP soil moisture is expected due to satellite orbit gaps and radio-frequency interference over parts of Bangladesh. The system automatically falls back to climatological soil moisture percentiles for affected district-days, with the fallback clearly disclosed in API responses via the `data_quality` field.

6.2 Query Testing

End-to-end tests are defined in `tests/test_e2e.py` and cover 7 scenarios including English and Bengali queries, alert triggering, crop advisory, abstention behaviour, and multi-turn session persistence.

Table 13 End-to-End Test Scenarios

ID	Scenario	Key Assertion
E2E-001	English drought query (Rajshahi)	SPI term in response; location detected; latency < 8s
E2E-002	Bengali drought query (Rajshahi)	Bengali response; <code>has_indicators=true</code>
E2E-003	Flood warning (Sylhet)	Water level cited; <code>alerts[]</code> populated
E2E-004	Crop advisory (Manikganj, Boro rice)	Phenological stage detected; crop calendar referenced
E2E-005	Out-of-scope abstention	System abstains; no hallucination; empty <code>retrieved_docs</code>
E2E-006	Bengali weather query (Dhaka)	Bengali response; Dhaka detected
E2E-007	Multi-turn session persistence	District context maintained across turns

6.3 Performance Metrics

Table 14 System Performance Benchmarks (Single T4 GPU, Prototype)

Metric	Value	Measurement Condition
End-to-end latency (p50)	< 5 seconds	1 concurrent user, English query
End-to-end latency (p95)	< 8 seconds	1 concurrent user, Bengali query (with translation)
Embedding throughput (ingestion)	~200 chunks/min	TEI batch size 32, T4 GPU
vLLM inference throughput	~50 tokens/sec	Granite-3.3-2B, FP16, T4 GPU
ArangoDB vector search latency	< 100 ms	k=15, 1,983 documents, HNSW index
Document ingestion rate	~60 docs/min	LLM labeling, GPU Docling
Total ingestion time (1,983 docs)	~33 minutes	LLM labeling strategy
Concurrent users supported	5	Without response quality degradation (<15s p95)
System uptime (with fallback)	> 99%	Including graceful-degradation paths

Common Failure Modes and Fixes

Table 15 Troubleshooting Reference

Symptom	Cause	Fix
vLLM CUDA out-of-memory	Model too large for T4 VRAM	Set <code>VLLM_GPU_MEMORY_UTILIZATION=0.80</code> ; reduce <code>VLLM_MAX_MODEL_LEN=4096</code>
GRAPH_TEST not found error	Graph not initialized before ingestion	Run ArangoDB shell graph creation script (Section 5.4)
Retriever returns empty docs	Score threshold too high; graph not populated	Lower <code>score_threshold</code> to 0.3; verify chunk count in ArangoDB
Dataprep POST 404	Endpoint path mismatch	Use <code>/v1/dataprep/ingest_files</code> (not <code>/v1/dataprep</code>)
Bengali text garbled	Wrong file encoding	Ensure UTF-8: <code>file -i document.md</code> ; convert with <code>iconv -t utf-8</code>
BMD scraper HTTP 403	BMD site down or rate-limited	Use cached data fallback: <code>bmd_scraper.py --use-cache</code>
SPI computation NaN	Insufficient data for gamma fit	Require min 20 years data; use <code>climate-indices</code> package as alternative

7. Appendices

A. API Endpoints Reference

The system exposes two API layers: the GENIE.AI internal service mesh (Docker network only) and the public FastAPI climate backend accessible via Kong gateway.

Public API (via Kong Gateway, port 443)

Table A-1 Public API Endpoints

Method	Endpoint	Auth	Description
POST	/api/auth/login	None	Authenticate; returns JWT token
POST	/api/chat	Bearer JWT	Submit query; returns advisory response with indicators and alerts
GET	/api/chat/sessions/{session_id}	Bearer JWT	Retrieve conversation history for session
POST	/api/documents/ingest	Admin JWT	Upload document for ingestion into knowledge base
GET	/api/documents	Admin JWT	List ingested documents with metadata
DELETE	/api/documents/{doc_id}	Admin JWT	Retract document from knowledge base
GET	/api/knowledge-hierarchy	Bearer JWT	Retrieve 2-level knowledge hierarchy JSON
POST	/api/knowledge-hierarchy	Admin JWT	Upload or update knowledge hierarchy
GET	/health	None	System health check; returns service status
POST	/webhook/whatsapp	Twilio sig.	WhatsApp message webhook (Twilio TwiML)

Climate Backend API (FastAPI, port 5001)

Table A-2 Climate Backend API Endpoints

Method	Endpoint	Description	Response Fields
GET	/health	Health check with data source status	status , data_sources , districts

POST	/chat	Full advisory query with indicator enrichment	response , session_id , location , indicators , alerts[] , language , data_quality
GET	/indicators/{district}	Current climate indicators for a district	spi_1 , spi_3 , spi_6 , ndvi_anomaly , sm_percentile , vhi , rainfall_anomaly_pct
GET	/districts	List all 64 supported districts with metadata	district_id , name , division , lat , lon
GET	/alerts	Active alerts across all districts	alerts[] with district , severity , indicator , value , message

GENIE.AI Internal Service Mesh

Table A-3 Internal Service Endpoints

Service	Host:Port	Key Endpoint
vLLM Inference	vllm:8000	POST /v1/chat/completions (OpenAI-compatible)
TEI Embeddings	tei:80	POST /v1/embeddings
TEI Reranker	tei_reranker:80	POST /rerank
Retriever	retriever-arango-service:7025	POST /v1/retrieval
Dataprep	dataprep-arango-service:5000	POST /v1/dataprep/ingest_files
ChatQnA Backend	chatqna-xeon-backend-server:8888	POST /v1/chatqna
ArangoDB	arango-vector-db:8529	HTTP REST API; AQL queries
Kong Admin	127.0.0.1:8001	GET /services , GET /routes

BMD AIS API Reference

Table A-4 Bangladesh Meteorological Department AIS API

Function Type	Key Parameters	Returns
get_bmdais_observation	station_no , data_date , data_hour	Temperature, humidity, wind, rainfall per station-hour
get_bmdais_forecast_product	product_date , product_type=weather	URL to national forecast PDF
get_bmdais_warning_product	product_date , product_type=warning	Active weather warnings and advisories

B. ArangoDB Schema

GRAPH_TEST_SOURCE Document Schema

```
// GRAPH_TEST_SOURCE vertex document schema
{
  "_key": "chirps_rajshahi_2026_q1",
  "_id": "GRAPH_TEST_SOURCE/chirps_rajshahi_2026_q1",
  "title": "CHIRPS Rainfall Report: Rajshahi District Q1 2026",
  "text": "Full document text content...",
  "category": "Climate & Weather Data",
  "subcategory": "Rainfall & Precipitation",
  "source": "gee_chirps_pipeline",
  "district": "Rajshahi",
  "division": "Rajshahi",
  "date_range": "2026-01-01/2026-03-31",
  "embedding": [0.023, -0.041, ...], // 768-dimensional float vector
  "chunk_count": 4,
  "ingested_at": "2026-05-10T12:34:56Z",
  "language": "en"
}
```

chunk Document Schema

```
// chunk vertex document schema
{
  "_key": "chirps_rajshahi_2026_q1_chunk_0",
  "_id": "chunk/chirps_rajshahi_2026_q1_chunk_0",
  "text": "Rainfall analysis for Rajshahi district (January-March 2026):
    Total precipitation: 42.3mm (87% of climatological normal).
    SPI-3 value: -0.23 (Near Normal). No drought conditions detected...",
  "embedding": [0.011, -0.088, ...], // 768-dimensional float vector
  "category": "Climate & Weather Data",
  "subcategory": "Rainfall & Precipitation",
  "chunk_index": 0,
  "token_count": 487,
  "source_doc_key": "chirps_rajshahi_2026_q1",
  "district": "Rajshahi",
  "metadata": {
    "date_range": "2026-01-01/2026-03-31",
    "spi_3_value": -0.23
  }
}
```

has_chunk Edge Schema

```
// has_chunk edge document schema
{
  "_key": "src_chirps_rajshahi_chunk_0",
  "_from": "GRAPH_TEST_SOURCE/chirps_rajshahi_2026_q1",
  "_to": "chunk/chirps_rajshahi_2026_q1_chunk_0",
  "chunk_order": 0,
  "created_at": "2026-05-10T12:35:00Z"
}
```

AQL Query Examples

```
// 1. Find all chunks for a specific district and category
FOR c IN chunk
  FILTER c.district == "Rajshahi"
    AND c.category == "Climate & Weather Data"
  RETURN c.text

// 2. Vector similarity search (top 5)
FOR doc IN chunk_view
  SEARCH APPROX_NEAR(doc.embedding, @query_embedding, 'cosine')
  SORT COSINE_DISTANCE(doc.embedding, @query_embedding) ASC
  LIMIT 5
  RETURN {text: doc.text, category: doc.category, score: 1}

// 3. Graph traversal from a source document
FOR v, e, p IN 1..1 OUTBOUND "GRAPH_TEST_SOURCE/chirps_rajshahi_2026_q1"
  GRAPH "GRAPH_TEST"
  RETURN v.text

// 4. Count documents by category
FOR doc IN GRAPH_TEST_SOURCE
  COLLECT category = doc.category WITH COUNT INTO n
  SORT n DESC
  RETURN {category, count: n}
```

Knowledge Hierarchy JSON Structure

```
// config/knowledge_hierarchy.json (abbreviated)
{
  "hierarchy": {
    "name": "Bangladesh Extreme Weather Advisor",
    "children": [
      {
        "name": "Climate & Weather Data",
        "children": [
          {"name": "Rainfall & Precipitation"},
          {"name": "Temperature & Drought Indices"},
          {"name": "Soil Moisture & Vegetation"}
        ]
      },
      {
        "name": "Agriculture & Food Security",
        "children": [
          {"name": "Crop Calendar & Phenology"},
          {"name": "Crop Production Statistics"},
          {"name": "Pest & Disease Alerts"}
        ]
      },
      {
        "name": "Policy & Guidelines",
        "children": [
          {"name": "Disaster Management"},
          {"name": "Agricultural Policy"},
          {"name": "Water Policy"}
        ]
      },
      {
        "name": "Research & Reports",
        "children": [
          {"name": "General Reference"}
        ]
      }
    ]
  }
}
```

C. Sample Queries & Responses

Sample 1: Drought Advisory (English)

```
// Request
POST /api/chat
{
  "message": "What is the current drought risk in Rajshahi district?",
  "session_id": "demo-001"
}

// Response (abbreviated)
{
  "response": "Based on current climate data for Rajshahi district (Rajshahi Division): The SPI-3 (3-month Standardized Precipitation Index) stands at -1.62, indicating SEVERE DROUGHT conditions. Rainfall from February to April 2026 was 38% below the climatological normal. NDVI anomaly is -0.14, showing moderate vegetation stress in croplands. Soil moisture is at the 12th percentile, critically below the 15th-percentile warning threshold.

  RECOMMENDED ACTIONS:
  1. Irrigate Boro rice (currently in tillering stage) if possible.
  2. Consider drought-tolerant variety substitution for late Aus planting.
  3. Monitor FFWC water level bulletins for river availability.

  Source: CHIRPS District Report Q1 2026; SPI Analysis May 2026;
  BAMIS Advisory Bulletin #14",
  "session_id": "demo-001",
  "location": {
    "district": "Rajshahi", "division": "Rajshahi",
    "confidence": 0.95, "lat": 24.37, "lon": 88.60
  },
  "indicators": {
    "spi_3": {"value": -1.62, "classification": "Severe Drought",
      "display_name": "SPI-3"},
    "ndvi_anomaly": {"value": -0.14, "classification": "Moderate Stress"},
    "sm_percentile": {"value": 0.12, "classification": "Critical"}
  },
  "alerts": [
    {"severity": "WARNING", "indicator": "spi_3",
      "message": "SPI-3 = -1.62 (Severe Drought) in Rajshahi"},
    {"severity": "WARNING", "indicator": "sm_percentile",
      "message": "Soil moisture at 12th percentile (critically low)"}
  ],
  "language": "en",
  "latency_ms": 4218
}
```

Sample 2: Bengali Query with Translation

```
// Request
POST /api/chat
{
  "message": "রাজশাহীতে খরার ঝুঁকি কেমন?",
  "session_id": "demo-002"
}

// Response (abbreviated)
{
  "response": "রাজশাহী জেলার বর্তমান জলবায়ু তথ্য অনুযায়ী: SPI-3  
(৩-মাসের মানসম্মত বৃষ্টিপাত সূচক) -১.৬২, যা তীব্র খরার ইঙ্গিত দেয়।  
ফেব্রুয়ারি থেকে এপ্রিল ২০২৬ পর্যন্ত বৃষ্টিপাত স্বাভাবিকের তুলনায়  
৩৮% কম ছিল। মাটির আর্দ্রতা ১২তম শতাংশে রয়েছে...",
  "language": "bn",
  "location": {"district": "Rajshahi", "confidence": 0.95}
}
```

Sample 3: Flood Warning Query

```
// Request
POST /api/chat
{
  "message": "Is there flood risk in Sylhet today?",
  "session_id": "demo-003"
}

// Response (abbreviated)
{
  "response": "For Sylhet district (Sylhet Division): Current BWDB/FFWC water  
level data shows the Surma River at Kanaighat station is 0.3m below danger  
level. Rainfall in the past 72 hours has been 45mm (above average for the  
season). SPI-1 is +0.8 (slightly wet).  
  
FLOOD RISK ASSESSMENT: MODERATE - Monitor closely. No active flood warning  
issued by FFWC as of today. Check FFWC daily bulletin for updates.",
  "alerts": [
    {"severity": "WATCH", "indicator": "water_level",  
"message": "Surma River approaching warning level at Kanaighat"}
  ],
  "language": "en"
}
```

Sample 4: Abstention (Out-of-Scope Query)

```
// Request - out-of-domain query
POST /api/chat
{
  "message": "What is the GDP of Bangladesh in 2025?",
  "session_id": "demo-004"
}

// Response
{
  "response": "I'm sorry, but I don't have relevant information about Bangladesh's GDP in my knowledge base. My expertise covers extreme weather advisories, drought monitoring, flood risk, and agricultural guidance for Bangladesh's 64 districts. For economic statistics, please consult the Bangladesh Bureau of Statistics or World Bank data portals.",
  "retrieved_docs": [],
  "language": "en"
}
// Note: CHATQNA_ENFORCE_ABSTENTION=true prevents hallucination
```